



PhishGuard

AI Email Phishing Detector with **Explainable Verdicts**

FutureTech HackFest 2026 · Applied AI × Cybersecurity · Chiara De Venuto

offline-firstprivacy-preserving98.75% CV accuracy

1 · The Problem

- Phishing is the #1 way attacks get started, with millions of new phishing sites appearing every quarter.
- The question that matters is not "is this link blacklisted?" but "**is this email trustworthy?**", and it needs an answer before you click.
- Most detectors are **black boxes**: they return a verdict with no reasons, so you never learn how to protect yourself.
- Most detectors also **depend on the cloud**, which means your email content is sent to a server somewhere else.

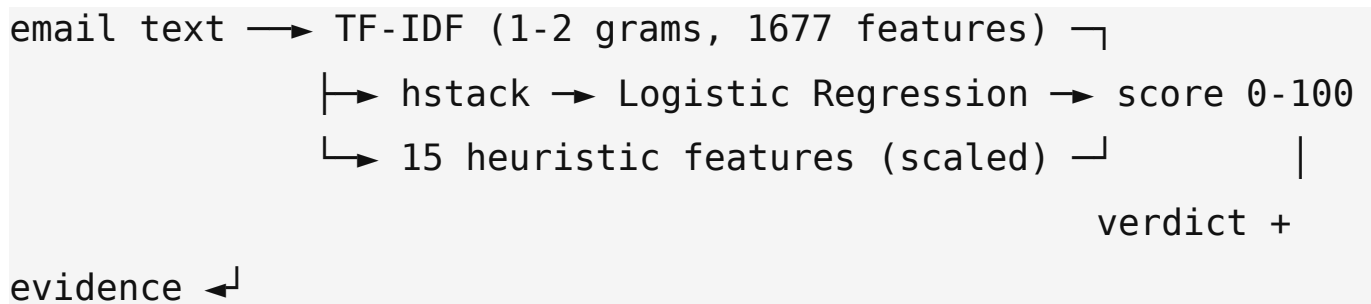
2 · Our Solution

You paste an email into PhishGuard and get:

- 🎯 **A risk score from 0 to 100** and a clear verdict: SAFE, SUSPICIOUS or PHISHING
- 📄 **Evidence you can read**: the exact signals and words that raised the score, so you understand the *why*
- 🗝️ **100% offline**: no API keys, no cloud, no data ever leaving your machine

- ⚡ A simple web interface, plus a JSON API for developers

3 • How It Works



- **ML layer:** picks up subtle patterns in the text that are hard to describe by hand.
- **Heuristic layer:** encodes expert knowledge such as typosquatting domains, raw-IP URLs, Reply-To mismatches, urgency keywords and attachment hints.
- **Interpretable by design:** logistic regression coefficients can be read directly, and the interface shows them to you.

4 • Data & Methodology

- **720 emails**, balanced between 360 phishing and 360 legitimate, generated deterministically (seed 42) from documented phishing patterns. The generator lives in the repository, so the dataset is **fully reproducible**.
- **Hard examples on purpose:** legitimate emails that mention words like "verify your email" or "password changed", plus low-intensity phishing that stays calm.
- Features: TF-IDF (1-2 grams) plus **15 interpretable heuristics**.
- Model: **Logistic Regression** with balanced classes. A strong baseline

that is fast and explainable.

- Evaluation: stratified holdout, **5-fold cross-validation** and an ablation study.

5 • Results

Metric	Holdout (20%)	5-fold CV
Accuracy	98.61%	98.75%
Precision	100%	100%
Recall	97.22%	97.50%
F1	98.59%	98.73%

Holdout confusion matrix: $\begin{bmatrix} 72 & 0 \\ 2 & 70 \end{bmatrix}$. Zero false positives, 2 false negatives.

6 • Demo Flow

- 1 Paste any email, suspicious or normal
- 2 Get the **score bar** and verdict badge
- 3 Read the **evidence**: active features and top indicators
- 4 Act on the advice, before clicking anything

100 **PHISHING** → 5 **SAFE**

Real output from the running app (see demo video).

7 • Innovation

- 🧠 **Explainability is the feature:** using PhishGuard teaches you what to look for, and security awareness is the first line of defense.
- 🔒 **Offline-first and private:** email content never leaves the device, which is rare among modern detectors.
- 🔬 **Reproducible data:** the whole corpus is generated by auditable, committed code.
- 🛡️ **Dual-layer defense:** machine learning plus expert heuristics, and both layers are visible to the user.

8 • Limits & Roadmap

- The dataset is synthetic, so validation on real-world mail (Enron corpora and live phishing feeds) is the immediate next step.
- Text indicators are single tokens today; natural-language explanations are the follow-up.
- Logistic regression is a strong baseline, and a fine-tuned transformer is a possible upgrade if you accept the trade-off in size and interpretability.

Roadmap

- Browser extension and email-client plugin
- Threat-intel blacklist as a third layer
- Better explanations for the SUSPICIOUS band

9 • Stack

Python 3.12Flask 3.1scikit-learn 1.9TF-IDFLogistic Regressionjoblibunittest

Open-source only • No external services • MIT

GitHub: public repository with full code, dataset generator, tests and docs.

10 • AI Use Disclosure

This project was built **with the assistance of generative AI tools**, used for coding support, debugging and writing documentation. Every line of code was executed, tested and verified locally, and every metric you see in this deck comes from running the code itself.

Declared in compliance with:

- **Regulation (EU) 2024/1689 (EU AI Act), Article 50** (transparency obligations, applicable since 2 August 2026)
- **Italian Law No. 132/2025** on artificial intelligence (Article 13)

Full transparency towards evaluators, users and third parties.

Thank you

 **PhishGuard**. Know why. Click safely.

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